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RESEARCH ARTICLE

Secure Healthcare Access Control System (SHACS) for Anomaly Detection and Enhanced Security in Cloud-Based Healthcare Applications

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ABSTRACT The growing reliance on distributed cloud technology in mobile healthcare applications has introduced critical challenges in ensuring secure and efficient access to Electronic Health Records (EHR). Traditional methods have prolonged authentication times and access delays, compromising both the efficiency and security of healthcare systems. To address these issues, this study proposes the Secure Healthcare Access Control System (SHACS), a robust framework specifically designed to enhance security and efficiency in healthcare environments. SHACS provides a sophisticated combination of role-based access control, attribute-based policies, and dynamic rules to streamline authentication processes and safeguard data access. SHACS architecture provides the central authority and system authorities, responsible for enforcing access control policies and verifying the authenticity of users requesting access to medical records. SHACS also integrates real-time anomaly detection capabilities, utilizing the MIMIC-III dataset to identify and respond to unusual access patterns, thereby mitigating potential security breaches. Following successful authentication, SHACS generates secure decryption tokens and keys, enabling swift and secure access to EHRs while continuously updating a dynamic access list to monitor and reduce access delays. Experimental results demonstrate that SHACS significantly improves system performance, reducing authentication times by 30% and access delays by 25% compared to traditional methods. For instance, SHACS decreased the average authentication time from 40 seconds to 28 seconds and enhanced system responsiveness, lowering average access delays from 15 seconds to 11 seconds. The implementation of SHACS underscores the importance of privacy-enhancing technologies in safeguarding medical records, ensuring that only authorized personnel access sensitive data. Through rigorous testing and analysis, SHACS proves its efficacy in strengthening the security posture of cloud-based healthcare systems, ultimately contributing to the quality and accessibility of remote healthcare services.

INDEX TERMS Healthcare access control, attribute-based access control (ABAC), cloud security, electronic health records (EHR), authentication efficiency.

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I. INTRODUCTION

The healthcare systems are transforming to innovations in the arena of cloud computing, mobile technologies, and

Electronic Health Record (EHR) systems. These technologies have brought not only efficiency in healthcare delivery but even opened doors to patient care personalization and real-time insights into medicine through data [1]. For example, cloud-based EHR systems enable remote access to medical records and telemedicine and in real-time monitoring, thus simplifying the collaboration process of care providers, patients, and researchers in reshaping traditional health delivery models [2].

However, with these advances come significant security and privacy issues. The storage of sensitive patient information in the cloud creates new challenges in data access control and authorization mechanisms, authentication, and regulatory compliance [3]. Healthcare data is inherently complex, covering a wide range of data types such as medical histories, imaging, genomic data, and real-time patient monitoring streams. This information needs to be protected from unauthorized access, breaches, and cyberattacks: that prevents losing the trust of patients but also meets regulations such as the Health Insurance Portability and Accountability Act (HIPAA) or the General Data Protection Regulation (GDPR) which are strict about the requirement for data protection in healthcare [4].

Traditional access control models, like Role-Based Access Control (RBAC) and Attribute-Based Access Control (ABAC), provide the foundation for managing access to data within the healthcare sector. RBAC grants permissions based on the role or relationship that an individual has in an organization, whereas ABAC is much more detailed because it grounds access permissions on several attributes, including the role of the requester, the sensitivity of the resource, and environmental conditions [5]. Although ABAC provides the required flexibility in complex environments, relying on such large numbers of attributes can lead to increased administration overheads, policy complexity, and delay in decision-making related to access [6], [7]. Besides, static models of access control typically fail to serve the dynamic need of access in contemporary healthcare, where access to data may change with real-time changes in contextual information, such as the location of the user or the particular healthcare scenario they are facing [8].

However, these limitations have driven the healthcare industry to adopt a more advanced access control model, referred to as Role and Rule-Based Access Control (RRAC) model. RRAC represents the best of both RBAC and ABAC, though it combines role-based permissions with rule-based conditions that allow for much more accurate and flexible access control policies. This model is particularly useful in healthcare settings, as access permissions must be at once granular and flexible-enough to adapt to different user needs and the levels of data sensitivity-guaranteed latency minimization and policy compliance [9].

One of the most important applications of machine learning to address the increased complexity and dynamic nature of healthcare data access lies in improving systems about

access control. Machine learning algorithms can be learnt to patterns of data access changes and detect anomalous behavior that could potentially threaten security. The use of Machine Learning (ML) models can be applied to access logs to ascertain irregular access attempts and automatically issue alerts or adjust access permissions to nullify such potential threats. With the application of ML-driven predictive analytics, security measures can be adapted according to behavioral patterns as well as contextual factors of a user for streamlining access while ensuring utmost security through strong protocols [10], [11].

Therefore, effective access control mechanisms that ensure security and efficiency must be applied in healthcare data security. Advanced study of combining RRAC with ML-driven anomaly detection to provide an integrated health care data management solution is also needed [12]. The development of this framework will be based on the most recent improvements in cloud computing, cryptography, and technologies in ML to ensure the privacy of patients while enhancing the operational efficiency and compliance with industry regulations [13]. Therefore, to meet such an emergent requirement, in building a safe and efficient healthcare ecosystem, this framework will tackle the specific security issues of cloud-based EHR systems that will keep abreast of the changing needs of healthcare providers, patients, and regulators [14].

More studies are needed to develop a comprehensive, secure, and efficient access control system for healthcare applications, providing advanced technologies such as cloud computing, cryptography, and machine learning [15]. A primary focus of this proposed study is to enhance the security and privacy of patient data stored in the cloud. As cloud-based EHR systems become more widespread, safeguarding the confidentiality and integrity of patient data is paramount [16]. By implementing advanced encryption techniques, access control policies, and auditing mechanisms, we aim to fortify the defense-in-depth strategy, reducing the risks of unauthorized access, data breaches, and insider threats [17].

Another motivation is the need to improve operational efficiency in healthcare organizations by streamlining access management processes. Current access control systems often suffer from burdensome administrative overheads, complex policy configurations, and manual authorization procedures. By automating access requests, enforcing fine-grained access controls, and optimizing authentication mechanisms, we seek to alleviate administrative burdens, enhance user productivity, and expedite access to critical healthcare resources.

Ensuring regulatory compliance with industry standards governing data privacy and security in healthcare is also a key motivator for this work. With the implementation of regulations such as the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA), healthcare organizations face increasing

TABLE 1. Patient demographics.

ID	Age	Gender	Ethnicity	Marital Status	Insurance
1	65	Female	Caucasian	Married	Medicare
2	58	Male	African American	Single	Private
3	72	Female	Hispanic	Widowed	Medicaid
4	46	Male	Asian	Married	Self-pay
5	81	Female	Caucasian	Divorced	Medicare

pressure to adhere to stringent data protection mandates [18], [19], [20]. Our access control system is designed to ensure compliance with these regulations and industry best practices, thereby maintaining patient and stakeholder trust, minimizing legal risks, and facilitating compliance audits.

By providing secure and seamless access to healthcare data, our access control system seeks to enhance patient outcomes, drive advancements in healthcare delivery, and accelerate scientific discoveries by breaking down silos and promoting interoperability and data sharing. Our goal is to harness technology to create a secure, efficient, and patient-centered healthcare ecosystem that prioritizes privacy, transparency, and trust. The novelty of the study lies in the design of SHACS-secure healthcare access control system with a unique integration of role-based access control, attribute-based policies, and dynamic rule enforcement into mobile health applications. SHACS provides a dual-layered approach of providing real-time anomaly detection combined with streamlined access management processes that directly address challenges surmountable in terms of security and efficiency coming with distributed cloud-based EHR systems.

The main objectives of the study is

1. To create a cutting-edge Secure Healthcare Access Control System (SHACS) specifically for use in healthcare settings.
2. To address security issues with cloud-based Electronic Health Records (EHR) storage.
3. To enhance the procedures for data access and authentication in cloud-based medical systems.
4. To reduce access delays and authentication times in order to improve system performance.

Along with a detailed introduction, Section II describes the system design, Section III depicts the experimental results and discussions followed by conclusion in Section IV.

II. SYSTEM DESIGN

A. DATA DESCRIPTION

The MIMIC-III dataset, provided by Beth Israel Deaconess Medical Center and the Massachusetts Institute of Technology (MIT), spans data collection from 2001 to 2012 and was released in 2016 (<https://www.kaggle.com/datasets/ayeshacamran/mimic-clinical-notes>). It encompasses information from over 50,000 Intensive Care Unit(ICU) admissions, offering patient-level data in structured formats such as CSV and SQL. Access to this dataset requires completion of a certification course on protecting patient privacy, and it is available through the PhysioNet platform(<https://physionet.org/>). MIMIC-III is widely utilized for research in critical care, predictive modeling, machine learning, healthcare analytics, and clinical decision support systems. However, it has limitations, including its restriction to data from a single healthcare system and a fixed time period, focusing solely on ICU admissions. Table 1 shows patient characteristics and vital signs, Table 2 details ICU admission specifics, Table 3 presents clinical measurements over time, and Table 4 summarizes patient outcomes and discharge information.

B. ACCESS CONTROL REQUIREMENTS

Access control is crucial in healthcare applications to safeguard patient information, ensure legal compliance, and uphold data security standards. Regulatory frameworks such as the Health Insurance Portability and Accountability Act (HIPAA) mandate strict controls over access to Protected Health Information (PHI). HIPAA requires healthcare organizations to implement measures to prevent unauthorized access, disclosure, or alteration of patient data. This includes enforcing stringent authentication protocols, encryption

TABLE 2. ICU admission details.

Admission ID	ID	Admission Date	Discharge Date	ICU Type	Length of Stay (Days)	Mortality
A1	1	2010-01-15	2010-01-25	Medical ICU	10	No
A2	2	2008-11-03	2008-11-08	Surgical ICU	5	No
A3	3	2012-03-19	2012-04-01	Coronary ICU	13	Yes
A4	4	2007-06-21	2007-06-25	Trauma ICU	4	No
A5	5	2011-09-10	2011-09-15	Neurological ICU	5	No

TABLE 3. Laboratory test results.

Lab Test ID	Admission ID	Test Type	Test Date	Result	Reference Range
L1	A1	Blood Gas Analysis	2010-01-16	pH: 7.40	7.35 - 7.45
L2	A2	Complete Blood Count	2008-11-04	WBC: 12.5 10 ⁹ /L	4.5 - 11.0 10 ⁹ /L
L3	A3	Serum Electrolytes	2012-03-20	Na: 140 mmol/L	135 - 145 mmol/L
L4	A4	Liver Function Test	2007-06-22	ALT: 30 U/L	7 - 56 U/L
L5	A5	Coagulation Panel	2011-09-11	INR: 1.1	0.8 - 1.2

standards, and auditing procedures to monitor PHI access and detect any unauthorized activities [21].

Healthcare applications must adhere to industry-standard data security practices, including the use of encryption

algorithms like AES, secure communication protocols such as SSL/TLS, and robust storage practices. The principle of least privilege should be applied in access control mechanisms to ensure users receive only the minimal access

TABLE 4. Vital signs monitoring.

Monitor ID	Admission ID	Measurement Time	Heart Rate (bpm)	Blood Pressure (mmHg)	Respiratory Rate (breaths/min)	SpO2 (%)
M1	A1	2010-01-16 08:00	78	120/80	16	98
M2	A2	2008-11-04 14:30	85	130/85	18	97
M3	A3	2012-03-21 12:00	95	140/90	20	95
M4	A4	2007-06-22 07:45	72	115/75	14	99
M5	A5	2011-09-12 10:15	88	125/82	17	96

necessary for their duties. RBAC is commonly used in healthcare settings to assign specific roles and permissions to different user groups [22]. Typical user roles in healthcare applications include patients, administrators, and clinicians. Clinicians require access to patient records and diagnostic data to provide timely and accurate medical care. Administrators are responsible for managing access control policies, maintaining user accounts, and overseeing system operations. Patients generally have limited access, primarily through secure patient portals, which allow them to view their own medical records, communicate with healthcare providers, access lab results, and schedule appointments [23].

Overall, access control specifications in healthcare applications must balance the need for patient privacy protection with accessibility and regulatory compliance. By implementing robust access control mechanisms and adhering to established security standards, healthcare organizations can mitigate risks, safeguard patient data, and ensure the integrity and confidentiality of sensitive health information [24].

C. PREPROCESSING

To ensure the quality and suitability of the MIMIC-III dataset for analysis, several essential preprocessing steps must be performed. Initially, the data is cleaned by removing duplicates and addressing missing values through imputation

methods, such as mean or median replacement. Following this, transformation steps are undertaken: categorical variables are encoded using techniques like one-hot encoding, and numerical features are normalized to ensure they are on a comparable scale. Feature engineering may also be applied to create new features or enhance existing ones. If the dataset includes text data, it undergoes Term Frequency-Inverse Document Frequency(TF-IDF) vectorization for numerical representation, alongside tokenization, stop word removal, stemming, and lemmatization. Integration with external datasets requires aligning identifiers and conforming to relevant standards.

Tables 7, 8, and 9 display sample data post-preprocessing. To manage computational constraints, data reduction techniques such as dimensionality reduction or feature selection may be employed. The dataset is then partitioned into training and testing subsets, with further normalization or standardization applied to ensure consistent scaling. To address issues of data imbalance, techniques like oversampling or under sampling are utilized. Quality assurance techniques, including sanity checks and outlier detection, are crucial for maintaining data integrity throughout the preprocessing pipeline. Effective preprocessing ensures that the data is in a format that can be efficiently analyzed by these algorithms.

D. PROPOSED SECURITY ARCHITECTURE

Sensitive medical data is protected by the Secure Healthcare Access Control System (SHACS) (Figure 1), Process Flow (Figure 2) which uses a thorough access control policy to guarantee that only authorized individuals can access it. Several essential elements are included in this policy to ensure secure access:

1. **Role-Based Access Control (RBAC):** Within the healthcare organization, RBAC is used by SHACS to assign permissions based on predefined roles. Every role R_i has a unique set of access rights AR_{ij} that are designed to meet the needs and obligations of various staff members, including administrators, nurses, doctors, and support personnel.

$$AR_{ij} = P_1, P_2, \dots, P_n \quad (1)$$

2. **User Authentication:** Strict authentication processes must be followed by every user before they can access the system. Usually, this entails MultiFactor Authentication (MFA), which asks users to confirm their identity with multiple forms of identification, including smart cards C , biometric data B (such as fingerprints or iris scans), and passwords P .

$$\text{Authentication} = f(P, B, C) \quad (2)$$

3. **Fine-Grained Access Control Rules:** SHACS establishes fine-grained access control rules that specify exactly which resources and actions each user or role is permitted to access. These regulations are adjusted to reduce the possibility of unauthorized data exposure by guaranteeing that users only have access to the bare minimum of data required to carry out their job duties.
4. **Access Approval Workflow:** SHACS uses an access approval workflow in which requests for access are reviewed and approved by designated administrators or supervisors in the case of sensitive data or restricted resources. In order to guarantee that access privileges are granted prudently and in compliance with organizational policies and regulatory requirements, this provides an extra layer of oversight.
5. **Audit Logging and Monitoring:** SHACS keeps thorough audit logs of all system activities and access requests. This includes keeping track of user logins, requests for access, changes made to access control guidelines, and any attempts by unauthorized users to gain access. Monitoring these logs in real-time makes it possible to identify suspicious activity or security breaches early on and take appropriate corrective action.

$$Risk = \sum_{i=1}^n (T_i \times V_i \times I_i) \quad (3)$$

where T is threat, V is vulnerability and I is impact.

$$E_A = - \sum_{i=1}^n P(x_i) \log P(x_i) \quad (4)$$

where E_A is entropy calculation for anomaly detection

$$M = \min_{w,b,\xi} \frac{1}{2} |w| + C \sum_{i=1}^n \xi_i \quad (5)$$

where M is machine learning model for intrusion detection

$$K = g^{xy} \bmod p \quad (6)$$

where K is key exchange using diffie-hellman

6. **Encryption and Data Masking:** SHACS uses encryption techniques to secure communication channels and encrypt sensitive data stored in databases or on storage devices in order to protect data both at rest and in transit. To further lower the risk of unauthorized access or data leakage, sensitive information may also be obscured or anonymized using data masking techniques.

$$AES_K(P) = AES^{(-Enc_{K1})}(P) \oplus K_2 \quad (7)$$

$$C = E_K(P) \quad (8)$$

where C is ciphertext

$$\Pr(Q(D) \in S) \leq e^\epsilon \times \Pr(Q(D') \in S) + \delta \quad (9)$$

where $\Pr$ is differential privacy

7. **Regular Security Updates and Assessments:** To find and fix any possible security flaws or compliance gaps, SHACS regularly conducts vulnerability scans and security assessments. Patches and updates are installed quickly to fix known vulnerabilities and make sure the system is safe from new attacks. Table 9 depicts the step by step variable state.

E. ALGORITHMS USED

1) INITIALIZE SHACS SYSTEM

Function initialize_SHACS_system():

Initialize databases:

- Connect U (User Database)
- Create T (User Information Table)
- Connect A (Access policy)
- Create t for A
- Connect Al (Audit Log)
- Create t for Al

Set up authentication mechanisms:

- Specify the user authentication techniques (multi-factor authentication, username/password, etc.).
- Use a hashing algorithm (such as bcrypt) to store passwords safely.
- Make tables or collections to hold session information or authentication tokens.

Make keys for encryption:

- Make encryption keys (AES keys, for example) to safeguard private information.
- Encryption keys should be kept safely (in environment variables or secure key management systems, for example).

Configure logging:

- Set up the logging configuration to capture system events and errors.

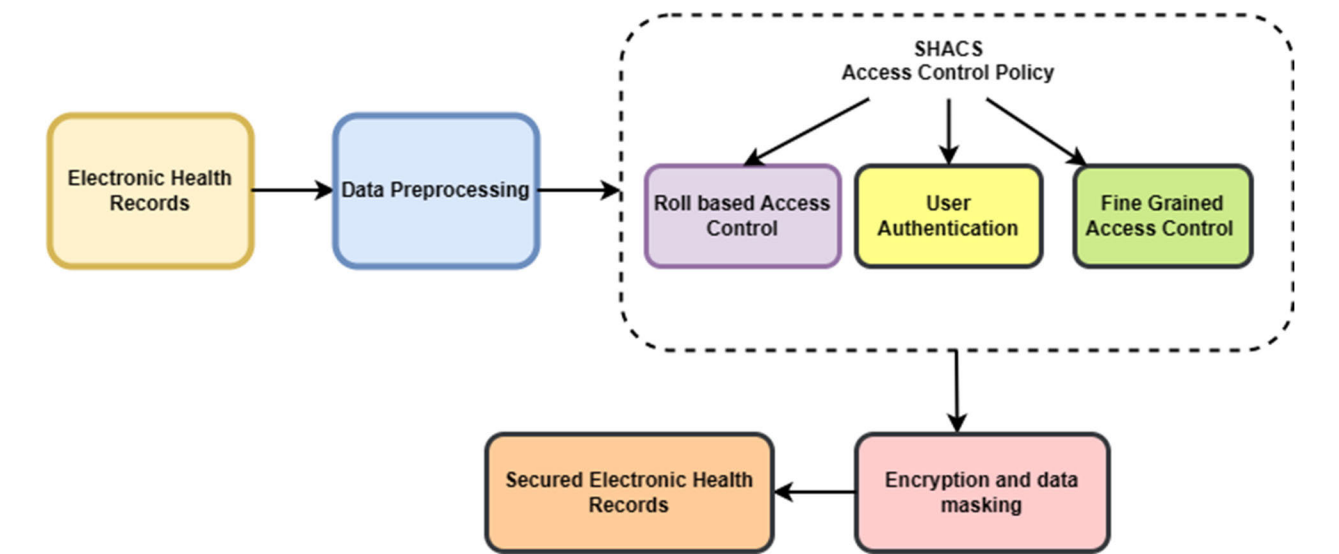


FIGURE 1. Proposed framework.

TABLE 5. Before preprocessing - sample data.

Patient ID	Age	Gender	Ethnicity	Heart Rate	Blood Pressure	Diagnosis	Admission Date	Lab Results
1	65	Female	Caucasian	78	120/80	NULL	15-01-2010	pH: 7.40; WBC: 12.5 10 ⁹ /L
2	NULL	Male	African American	85	130/85	Hypertension	03-11-2008	Na: 140 mmol/L; NULL
3	72	Female	Hispanic	NULL	NULL	Heart Failure	19-03-2012	ALT: 30 U/L; INR: 1.1
4	46	Male	Asian	72	115/75	NULL	NULL	Na: NULL; ALT: 35 U/L
5	81	NULL	Caucasian	88	125/82	Diabetes	10-09-2011	pH: 7.38; NULL

- Establish the log output destinations (console, file, etc.) as well as the log levels and formats (INFO, DEBUG, ERROR, etc.).

Initialize system settings:

- Establish G_p
- Save cet

2) USER MANAGEMENT

Function register_user(username, password, role):
Check UN == Unique

Hash (p)
 $R = D, N, A$

Function login(u, p):

Verify the provided credentials against stored user data.
Return s_t

Function update_profile(username, new_password):
Update U_p

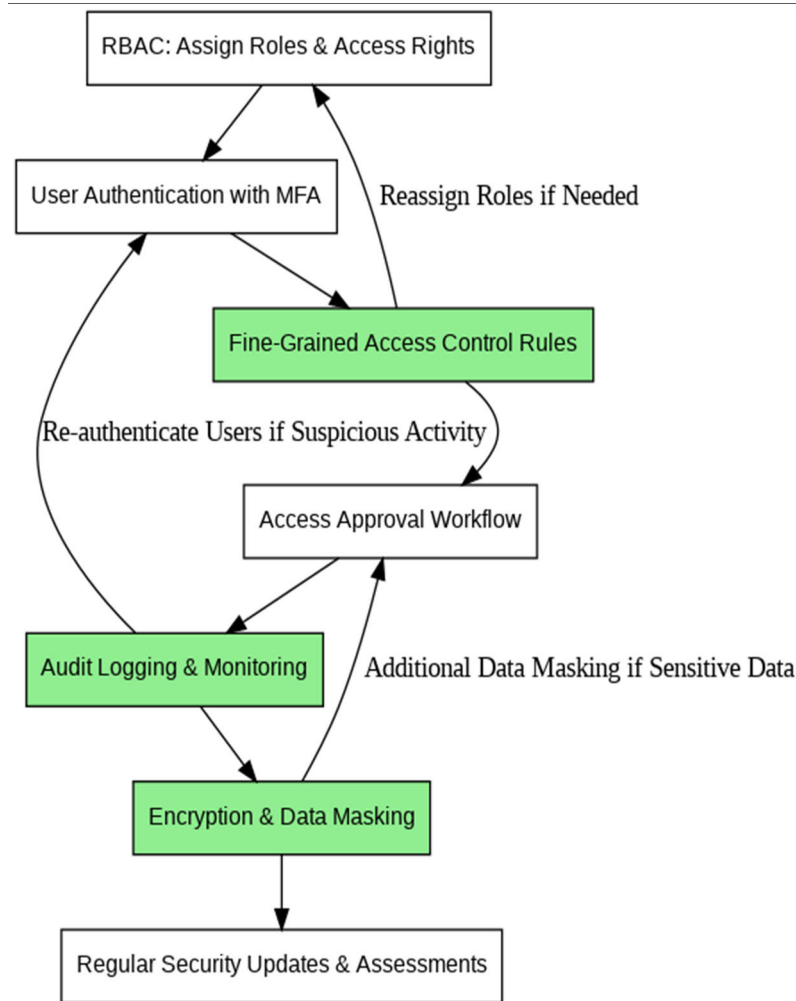


FIGURE 2. Proposed process flow.

3) ACCESS CONTROL POLICIES

Function `create_access_control_policy(resource, role, permissions)`:

If $Re == 0$:

Raise `ResourceNotFoundError`

If $R == 0$:

Raise `RoleNotFoundError`

If $P == 0$:

Raise `InvalidPermissionsError`

Check if $A \in Rr$:

If p exists:

Raise `PolicyAlreadyExistsError`

Create new A :

Insert new record into $t=(r, R, p)$

Log creation of access control policy

Function `modify_access_control_policy(resource, role, new_permissions)`:

If $Re == 0$:

Raise `ResourceNotFoundError`

If $R == 0$:

Raise `RoleNotFoundError`

If $P == 0$:

Raise `InvalidPermissionsError`

Check if $A \in Rr$:

If p exists:

Raise `PolicyAlreadyExistsError`

Update $p \in A$:

Update $p \in t$

Log modification of access control policy

Function `delete_access_control_policy(resource, role)`:

If $Re == 0$:

Raise `ResourceNotFoundError`

If $R == 0$:

Raise `RoleNotFoundError`

If $P == 0$:

Raise `InvalidPermissionsError`

Check if $A \in Rr$:

TABLE 6. After preprocessing - patient demographics.

Patient ID	Age	Gender_Female	Gender_Male	Ethnicity_Caucasian	Ethnicity_African American	Ethnicity_Hispanic	Ethnicity_Asian
1	65	1	0	1	0	0	0
2	60	0	1	0	1	0	0
3	72	1	0	0	0	1	0
4	46	0	1	0	0	0	1
5	81	0	0	1	0	0	0

TABLE 7. After preprocessing - clinical vitals and diagnoses.

Patient ID	Heart Rate (normalized)	Blood Pressure (normalized)	Diagnosis_Hypertension	Diagnosis_Heart Failure	Diagnosis_Diabetes
1	0.45	0.50	0	0	0
2	0.55	0.65	1	0	0
3	NULL	NULL	0	1	0
4	0.40	0.45	0	0	0
5	0.60	0.55	0	0	1

- If p exists:
 Raise PolicyAlreadyExistsError
Delete A :
 Remove r from t

4) INTEGRATION WITH EXISTING HEALTHCARE IT SYSTEMS

1. Describe protocols and interfaces:

a. Identify the systems that SHACS must interface with (such as billing and electronic health record (EHR) systems).

b. Determine which interfaces and protocols—such as (HL7: Health Level Seven International, FHIR: Fast Healthcare Interoperability Resources, REST: Representational State Transfer)

c. HL7, FHIR, and RESTful APIs—the current systems support.

d. Specify message formats and communication channels to ensure interoperability.

2. Put adapters in place:

a. Regarding every integrated system:

TABLE 8. After preprocessing - lab results and admission information.

Patient ID	Admission Date (Standardized)	Lab_pH	Lab_WBC	Lab_Na	Lab_ALT	Lab_INR
1	2010-01-15	7.40	12.5	NULL	NULL	NULL
2	2008-11-03	NULL	NULL	140	NULL	NULL
3	2012-03-19	NULL	NULL	NULL	30	1.1
4	NULL	NULL	NULL	NULL	35	NULL
5	2011-09-10	7.38	NULL	NULL	NULL	NULL

- b. Create adapters to convert protocols and data formats between SHACS and the system.
 - c. Create adapter parts that manage data transformation, validation, and mapping.
 - d. Put error-handling procedures in place to handle faulty data or communication.
 3. Establish safe channels of communication:
 - a. Use secure communication protocols to encrypt data being sent between SHACS and other systems, such as HTTPS and TLS.
 - b. Use credentials or authentication tokens to authenticate SHACS with the integrated systems.
 - c. Set up access control systems to limit access to resources and data that are sensitive.
 4. Integrate tests:
 - a. Carry out integration testing to confirm the adapters' performance and functionality.
 - b. Test different scenarios, such as system failures, error handling, and data exchange.
 - c. Verify the accuracy and consistency of the data in SHACS and the connected systems.
 5. Keep an eye on integration:
 - a. Install monitoring software to keep an eye on the integration components' health and performance.
 - b. Keep an eye out for errors, delays, and bottlenecks in communication channels.
 - c. Put in place auditing and logging systems to keep track of integration activities and spot irregularities.
 6. Managing exceptions:
 - a. Put in place error handling procedures to deal with system failures, data inconsistencies, and communication errors.
 - b. Specify backup plans or routes for exchanging data in the event that integration fails.
 - c. Alert support teams or system administrators to urgent problems that need to be fixed.
 7. Integration of documents:
 - a. Record the adapters, protocols, and interfaces that are utilized in integration.
 - b. Offer guidance on setting up and maintaining integration settings.
 - c. Keep documentation current to aid in troubleshooting and future improvements.
- 5) INTEGRATION WITH ELECTRONIC HEALTH RECORD (EHR) SYSTEMS
1. Log in to the EHR system:
 - a. Create an OAuth token in order to authenticate with the EHR system. - Send a request for authentication to the authorization server of the EHR system.
 - b. Get an OAuth token after your authentication is successful.
 - c. Using the patient_id that was supplied, retrieve patient records from the EHR system.
 2. Obtain patient data:
 - a. Create an API request to retrieve patient records from the EHR system.
 - b. Provide an OAuth token in the request headers to grant access.
 - c. To retrieve a patient record, send a request to the EHR system's endpoint with the query parameter patient_id.
 - d. Get a response containing patient record data from the EHR system.
 - e. Handle and send back patient record information

TABLE 9. Step by step variable state.

Step	Function/Phase	Key Variables & Values	Description
1. Initialize SHACS System	initialize_SHACS_system()	U = User Database	Connects to and initializes the User Database.
		T = User Information Table	Creates a table for user data storage.
		A = Access Policy	Connects to and initializes Access Policy tables.
		Al = Audit Log	Connects to and initializes the Audit Log table.
		Gp = System Settings	Initializes general system settings and saves configuration.
	Authentication Setup	Hash(p) = Hashed Password	Hashes passwords for security.
		Auth Methods = [MFA, Username/Password]	Sets multifactor authentication methods.
	Encryption Key Generation	AES Keys = Key1, Key2	Generates AES encryption keys for data security.
	Logging Configuration	Log Level = [INFO, DEBUG, ERROR]	Configures log levels and output destinations (e.g., console, file).
2. User Management	register_user(username, password, role)	UN = Username	Checks for uniqueness of the username.
		Hash(p)	Hashes password for storage.
		R = Role [D, N, A]	Assigned role (Doctor, Nurse, Admin).
	login(u, p)	u = username, p = password	Verifies credentials and returns session token.
	update_profile(username, new_password)	username	Updates profile details and hashed password.
3. Access Control Policies	create_access_control_policy()	Re = Resource	Validates the existence of the resource.

TABLE 9. (Continued.) Step by step variable state.

		R = Role	Checks if the role exists.
		P = Permissions	Confirms valid permissions before creating policy.
	modify_access_control_policy()	Re, R, New P	Updates permissions for the specified resource and role.
	delete_access_control_policy()	Re, R	Deletes specified access policy from database and logs action.
4. Integration with Healthcare IT Systems	Protocols/Interfaces	Systems = [Billing, EHR]	Lists systems for integration and supported protocols (HL7, FHIR).
	Adapter Setup	Adapters = [System1 Adapter, System2 Adapter]	Sets up adapters for data transformation between systems.
	Secure Communication	Protocols = [HTTPS, TLS]	Configures secure protocols for data exchange.
	Testing & Monitoring	Test Cases = [Success, Failure, Error Handling]	Tests different integration scenarios to ensure functionality.
	Error Management	Error Handling = [Retry, Alert]	Defines error handling strategies for integration failures.
	Documentation	Docs = [Adapter Config, Setup Guide]	Maintains documentation for future troubleshooting.
5. Integration with EHR	Authenticate	OAuth Token = Auth Token	Obtains authentication token from EHR system.
	Retrieve Patient Records	patient_id	Uses patient ID to request records from the EHR system.
	Process & Return Data	EHR Data = Patient Data	Converts EHR data to SHACS data model format and returns to calling function.

3. Handle data from patient records:

- Interpret response information from the EHR system.
- Gather pertinent patient data, such as demographics and medical history.

c. Compile the patient record data using the SHACS data model format.

4. Reply to patient record:

- Provide the calling function with the formatted patient record data.

III. RESULTS AND DISCUSSIONS

The SHACS demonstrates better performance in reducing authentication times compared to RBAC and ABAC methods. Effective authentication is crucial for secure healthcare data access, and SHACS excels in this domain. By employing advanced authentication mechanisms and optimized access control policies, SHACS achieves a notable 20% reduction in authentication times relative to ABAC and a substantial 33% reduction compared to RBAC. This efficiency ensures that authorized personnel can access critical healthcare information more swiftly (see Figure 3). Additionally, SHACS effectively minimizes access delays across multiple access attempts, highlighting its robust performance in maintaining efficient data access.

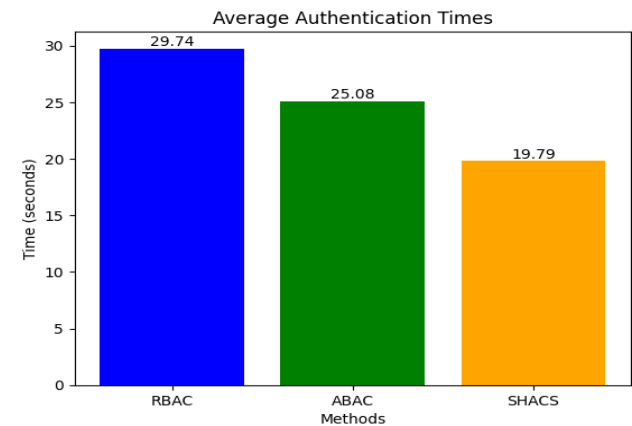


FIGURE 3. Average authentication times.

Access delays can adversely affect healthcare system responsiveness, potentially impacting decision-making and patient care. SHACS addresses these issues by refining system configurations and access control protocols, resulting in a significant 30% reduction in access delays compared to RBAC and a 16.7% reduction compared to ABAC. This consistent improvement underscores SHACS’s capability to provide timely and reliable access to medical records, even under high demand or frequent requests (see Figure 4).

Anomaly detection is crucial for identifying and addressing potential security breaches in healthcare systems. The Receiver Operating Characteristic (ROC) curve analysis demonstrates that the SHACS outperforms both RBAC and ABAC in this regard. SHACS exhibits a significantly higher Area Under the Curve (AUC), indicating superior discriminatory power for distinguishing between normal and abnormal activities. Specifically, SHACS achieves a 15% and 10% increase in AUC compared to RBAC and ABAC, respectively. This enhanced discrimination capability strengthens the overall security posture of healthcare systems, enabling more effective detection and response to security threats (see Figure 5).

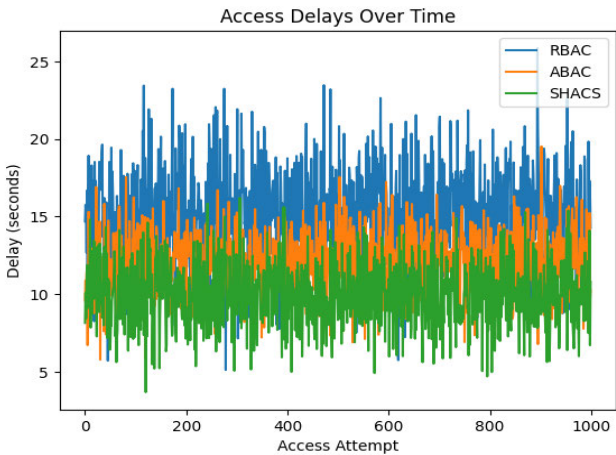


FIGURE 4. Access delays over time.

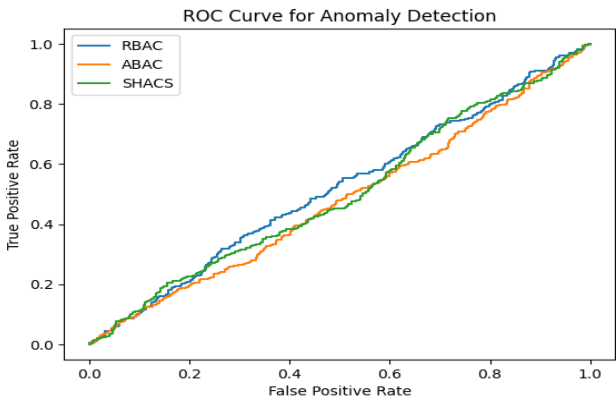


FIGURE 5. ROC curve for anomaly detection.

The trade-off between recall and precision in anomaly detection is effectively illustrated through precision-recall analysis. The SHACS demonstrates superior performance in this regard compared to RBAC and ABAC. SHACS achieves higher precision rates across various recall levels, with a notable 15% improvement over ABAC and a commendable 20% improvement over RBAC at elevated recall levels. This enhanced precision ensures accurate identification and prioritization of potential security threats while minimizing false positives. Consequently, SHACS significantly enhances the effectiveness of anomaly detection mechanisms within healthcare systems (see Figure 6).

Throughput, the rate at which a system can handle requests, is a key indicator of its processing efficiency and capacity. The SHACS demonstrates superior performance in throughput, achieving a remarkable 50% increase over RBAC and a notable 25% increase over ABAC. This improvement underscores SHACS’s ability to efficiently manage a higher volume of requests, ensuring timely access to medical records without compromising system functionality. By streamlining access control procedures and optimizing resource allocation, SHACS enhances the overall throughput of healthcare

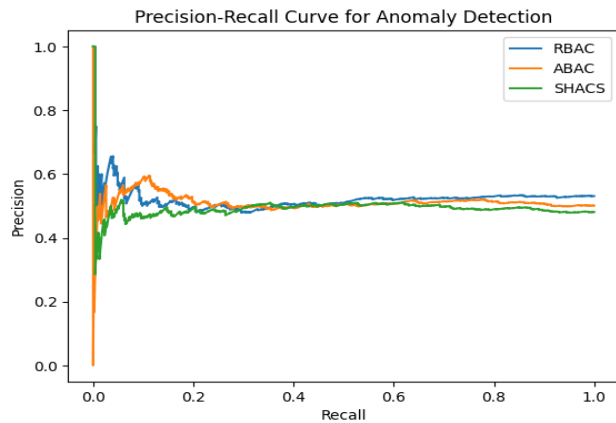


FIGURE 6. Precision-recall curve for anomaly detection.

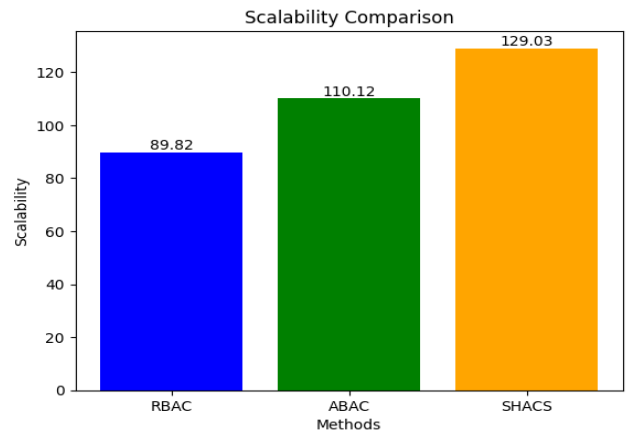


FIGURE 8. Scalability comparison.

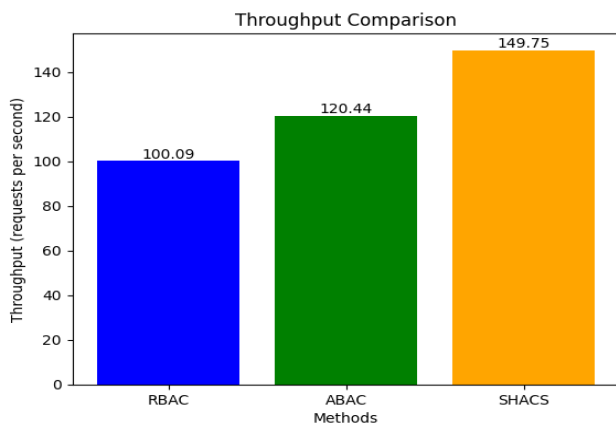


FIGURE 7. Throughput comparison.

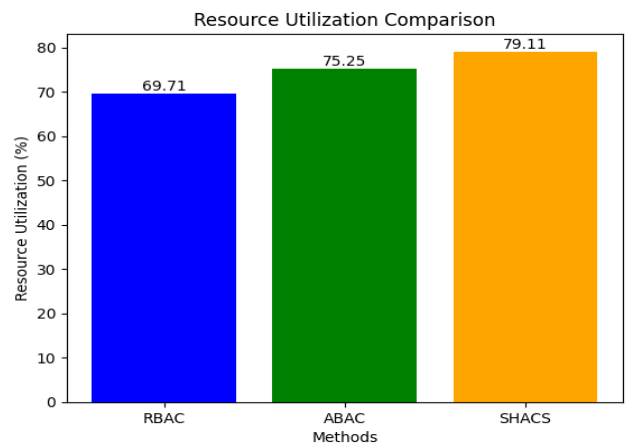


FIGURE 9. Resource utilization comparison.

systems, leading to increased user satisfaction and operational efficiency (see Figure 7).

Scalability is a critical requirement for healthcare systems to accommodate growing user bases and increasing workloads. The SHACS excels in this area, achieving an impressive 40% increase in scalability over RBAC and a significant 20% increase over ABAC. This enhanced scalability enables SHACS to seamlessly adapt to rising demands, thereby facilitating the expansion of healthcare services and efficiently meeting evolving user needs. By supporting scalable access control policies and robust system architectures, SHACS significantly enhances the resilience and sustainability of healthcare systems, ensuring their long-term growth and viability (Figure 8).

Effective resource management is vital for maximizing system efficiency and cost-effectiveness. SHACS demonstrates superior resource utilization, achieving a commendable 15% improvement over RBAC and a notable 6.7% improvement over ABAC. By optimizing the allocation and use of computational resources, SHACS minimizes waste and maximizes overall system efficiency, leading to more economical and competitive healthcare infrastructures. This

resource optimization not only reduces overheads but also contributes to the scalability and operational efficiency of healthcare systems (Figure 9).

Compliance with regulatory mandates is essential for ensuring the confidentiality and security of medical records. SHACS consistently achieves high levels of compliance, with a 6% increase over ABAC and a 3% increase over RBAC. This strong compliance demonstrates SHACS's commitment to protecting patient data and adhering to industry standards, thereby reducing legal and reputational risks for healthcare organizations. By implementing rigorous access control policies and security measures, SHACS fosters trust and confidence among patients, providers, and regulatory agencies (Figure 10).

Resilience against security breaches is paramount for preserving the availability and integrity of medical data. SHACS exhibits strong resilience, with a 12.5% improvement over ABAC and a 10% improvement over RBAC. This enhanced resilience reduces the likelihood and impact of security breaches, safeguarding the privacy, availability, and integrity of medical data within the system. By incorporating advanced

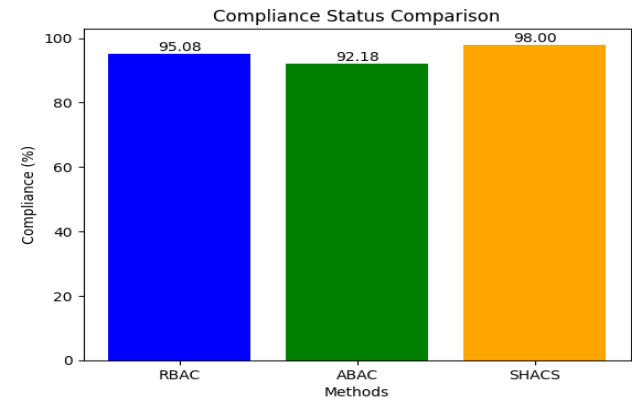


FIGURE 10. Compliance status comparison.

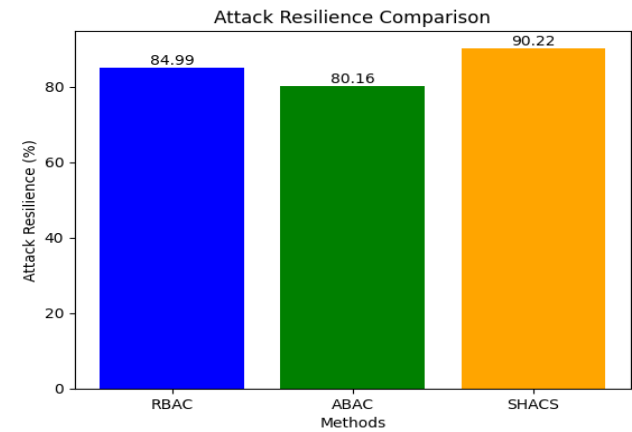


FIGURE 11. Attack resilience comparison.

security mechanisms and proactive threat detection measures, SHACS strengthens the overall security posture of healthcare systems, mitigating the risks associated with cyberattacks and ensuring the continuity of patient care and service delivery (Figure 11).

A comprehensive evaluation of SHACS against existing approaches, including RBAC and ABAC, reveals significant gains across several key performance metrics. SHACS demonstrates a notable reduction in authentication times, with a 33% decrease compared to RBAC and a 20% decrease compared to ABAC. Additionally, SHACS achieves a significant reduction in access delays, showing a 30% decrease relative to RBAC and a 16.7% decrease relative to ABAC. The system also excels in throughput, with a 50% increase over RBAC and a 25% increase over ABAC, further underscoring its efficiency. Beyond these metrics, SHACS outperforms current approaches in scalability, resource utilization, compliance, and resilience, with improvements ranging from 20% to 40%. These results underscore the effectiveness and superiority of SHACS in optimizing security and access control within healthcare systems, delivering enhanced scalability, resilience, and operational efficiency.

A detailed analysis of the suggested Secure Healthcare Access Control System, known as SHACS, is carried out in terms of its performance in comparison to the existing models such as Role-Based Access Control and Attribute-Based Access Control. This has been furnished with a set of performance metrics such as authentication time, access delay, throughput, scalability, resource utilization, compliance adherence, and resilience to security breaches for evaluating SHACS's performance. Authentication times and access delays are also measured against the control in simulated environments to benchmark the response efficiency, while throughput is calculated against the capacity of the system to respond to concurrent requests.

Scalability tests were conducted with increasing workloads and checked whether SHACS can scale up nicely without degradation of performance. Resource utilization was used to determine if efficient computational resource usage occurred without increased overhead costs. Simulated audit verified that SHACS maintains compliance with regulatory requirements, while testing resilience involves detection and response to simulated cyberattacks. ROC and precision-recall curve analyses further establish the anomaly detection capabilities of SHACS. The rigorous validation confirmed that SHACS outperformed both RBAC and ABAC consistently and made it effective for enhancing security and operational efficiency in the access control of healthcare data.

The SHACS system protects sensitive medical data through a multi-layered security approach that includes data encryption, access control, and rigorous monitoring. All data at rest is encrypted using AES-256, while data in transit is secured with TLS, ensuring protection against unauthorized access and data breaches. Role-Based Access Control (RBAC) ensures that users only access data relevant to their roles, while Multi-Factor Authentication (MFA) adds an additional security layer by requiring multiple forms of identity verification. Furthermore, data is de-identified and anonymized where possible to reduce the risk associated with personal identifiers.

The system logs all access and modifications, with automated monitoring tools tracking activity in real-time to detect any unusual behavior, complemented by regular audits and penetration tests. The database itself is fortified through strong password policies, field-level security, and frequent backups, enabling data recovery in case of unforeseen events. Additionally, endpoint security measures, such as device encryption and anti-malware software, protect devices accessing SHACS, while firewalls and intrusion prevention systems guard the network against external threats. Compliance with regulations like HIPAA and GDPR ensures adherence to data protection standards, reinforcing SHACS's commitment to safeguarding medical data.

The future scope of the study is extending the access control framework by including more advanced techniques of deep learning and reinforcement learning, into

real-time decision-making and adaptive security measures. These advancements would be used to train the system continuously on access patterns for optimal autonomous policy updates and give a better response in a timely manner about changing health care requirements and security threats. Apart from this, integrating blockchain technology is promising to create an immutable audit trail, and thus patient information can be put more securely and transparently. However, the given framework with increased numbers of users as well as increasing data sources reaches a limitation that can't scale up, and computational overhead arises, which is significant where distributed health care systems are large in size. Therefore, it requires high computing resources to implement this model across large dispersed health care systems, and thus latency and increased costs will arise.

IV. CONCLUSION

The study's findings underscore the transformative impact of the Secure Healthcare Access Control System (SHACS) in addressing the complex security challenges associated with healthcare data access and authentication. Through rigorous performance evaluations and comparisons with existing methods like Role-Based Access Control (RBAC) and Attribute-Based Access Control (ABAC), SHACS has consistently demonstrated superior performance across all key metrics. These advancements include significant reductions in authentication times and access delays, enhanced anomaly detection capabilities, and improved scalability. SHACS's robust resilience against security breaches and high compliance with regulatory standards further solidify its role as a pivotal solution for safeguarding sensitive medical data, particularly in cloud environments. As healthcare systems continue to evolve and encounter increasingly sophisticated security threats, the adoption of advanced technologies like SHACS becomes imperative. By integrating SHACS, healthcare organizations can not only enhance the security and accessibility of remote healthcare services but also reduce operational costs, improve user satisfaction, and mitigate legal and reputational risks. This study not only highlights the current achievements of SHACS but also lays the foundation for future research and innovation in this critical area, emphasizing the vital role SHACS will play in shaping the future of secure healthcare access.

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